

INTERPRETING LINEAR RELATIONSHIPS FROM TABLES AND WRITTEN DESCRIPTIONS



LESSON INTRODUCTION

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph or an equation in slope-intercept form.

LEARNING TARGETS

- I can determine and interpret the slope and y-intercept of a two-variable linear equation from a table of values in a real-world context.
- I can determine and interpret the slope and y-intercept of a two-variable linear equation from a written description in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

MA.912.AR.2.5 Solve and graph mathematical and real-world problems that are modeled with linear functions. Interpret key features and determine constraints in terms of the context.

ESSENTIAL QUESTIONS

- How do you determine and interpret the slope and y-intercept of a two-variable linear equation from a table of values in a real-world context?
- How do you determine and interpret the slope and y-intercept of a two-variable linear equation from a written description of a real-world context?

LESSON NARRATIVE

Students will determine and interpret the slope and y-intercept in context when given either a table of values or a written description. Students will use their knowledge of how to find slope using a table or points as well as determining the y-intercept using the slope found from two points. They will then interpret these values in each real-world context.

COMPONENT SUMMARY

7.7.1 WARM-UP (~5 MIN.)

Interpret the slope and y-intercept of an equation on a Florida-based context

7.7.3 GUIDED PRACTICE (10 MIN.)

Practice determining and interpreting the slope and y-intercept from tables and written descriptions in context with a partner

7.7.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of determining and interpreting the slope and y-intercept from a written description in context

7.7.2 GUIDED INSTRUCTION (10-15 MIN.)

Learn how to determine and interpret the slope and y-intercept from tables and written descriptions in context

7.7.4 YOUR TURN! (10 MIN.)

Independently practice determining and interpreting the slope and y-intercept from tables and written descriptions in context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Slope
- Rate of change
- y-intercept
- Linear relationship

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to find the y-intercept from a table of values when it is not included in the table.
- Students may have a difficult time finding the y-intercept in a written description if they do not understand it is the initial value in the situation.
- Students may struggle to interpret specific contexts if they are not familiar with the contexts.

THINGS TO CONSIDER

- Consider explicitly spiraling content by assigning students to calculate the slope using different values in the table to reiterate that it does not matter which two points are used.
- Consider allowing students to use a calculator to perform calculations. This will allow them to focus on interpreting the values.
- Provide students the opportunity to read and discuss each context prior to performing calculations and interpreting to ensure understanding of each context.

LITERACY AND LANGUAGE ROUTINES

Three Reads

7.7.2 Guided Instruction and 7.7.3 Guided practice provide opportunities to use this structure. Consider applying the Three Reads routine as students make sense of each unique context. This routine supports reading comprehension and supports students in negotiating what information is relevant from a text when discussing with their partner.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students will be asked to interpret slopes and y-intercepts of linear representations in real-world contexts. Given a table of values or a written description, students will interpret their findings according to the context. For example, in 7.7.5 Wrap-Up, students must determine the slope and y-intercept from the written description and then interpret their meanings to the context.

DIFFERENTIATE INSTRUCTION

Both 7.7.3 Guided Practice and the 7.7.4 Your Turn! components provide students the opportunities to determine and interpret the slope and y-intercept using both a table of values and a written description. Consider combining these two components and allowing students to choose which questions to complete. For example, there are two questions where the student is given a table of values. Allow the student to choose which one to complete. Consider engaging students with the Think-Pair-Share routine in 7.7.3 Guided Practice. The discussions will provide auditory entry points.

7.7.1 WARM-UP: LANGUAGES SPOKEN IN FLORIDA

ENRICHMENT

Consider asking students the following:

- Do you or anyone you know speak any languages other than English?
- What are some other languages you know are spoken by people living in Florida that are not listed?
- If you could learn another language, what language would you want to learn? Why?
- What are the benefits of being able to speak more than one language?



7.7.1 Warm-Up: Languages Spoken in Florida

Florida has the third largest population of all states within the U.S. While nearly $\frac{3}{4}$ of the population in Florida speaks English at home, there are actually more than 200 languages spoken in homes throughout the state. Some of these languages include Spanish, Haitian Creole, French, Portuguese, German, Tagalog, Arabic, and Mandarin.

The equation $y = 128x + 1018$ can be used to model the relationship between the number of people in Florida who speak a language other than English at home, y , in thousands, and the number of years since 1980, x .

1. Complete the statement to interpret this relationship in context.

In 1980, approximately

- 128
- 1,018
- 128,000
- 1,018,000

Floridians spoke a

language other than English at home. Since then, the

number of Floridians speaking a language other than

English at home has grown by approximately

- 128
- 1,018
- 128,000
- 1,018,000

people every year.



7.7.2 Guided Instruction: Interpreting Slope and y-intercept from Tables and Descriptions

1. Econ Rental Car is a company that rents cars using a linear pricing plan. The company charges a daily rate in addition to maintenance fees. The total cost in dollars, y , to rent a car for x days is shown in the table.

x	y
2	97
4	187
6	277
10	457

- a. Find the slope for this relationship. Show your work.

$$\frac{187-97}{4-2} = \frac{90}{2} = 45$$

- b. What does the slope mean in this context?

The daily rate the company charges is \$45 per day.

- c. Zoey noticed that the table does not include a row where $x = 0$. She said there is no way to know the y -intercept from the table. Discuss with your partner how the information in the table could be used to determine the y -intercept.

7.7.2 GUIDED INSTRUCTION: INTERPRETING SLOPE AND Y-INTERCEPT FROM TABLES AND DESCRIPTIONS

TEACHER MOVES

Give students the opportunity to read each real-world context on their own. Then consider using a Turn-and-Talk strategy to have them summarize what is happening in their own words with a partner. Allowing students to discuss and make meaning of the context first will set them up to be able to better interpret them on their own.

TEACHER MOVES

During the component, spiral questions about content from previous lessons to help students make connections between their previous knowledge and the new content. For example, when discussing how to find the slope from a table, help students remember that any two points can be used to find the slope. This can be done by assigning groups of students to use different sets of points and then discussing why they all arrived at the same answer.

TEACHER MOVES

Listen as students discuss their ideas for how they might determine the y -intercept from the table. Revisit any unique ideas after completing question 2 and allow students to reflect on which student, if either, in question 2 used similar thinking as they discussed with their partner.

7.7.2 GUIDED INSTRUCTION: INTERPRETING SLOPE AND Y-INTERCEPT FROM TABLES AND DESCRIPTIONS (CONTINUED)

ENRICHMENT

Consider asking students the following:

- Explain which method makes the most sense to you. Is either method similar to what you discussed with your partner?
- How did you use the original context to interpret the meaning of the y-intercept?

2. Carson and Jordyn determined the y -intercept using different reasoning. Their thinking is shared in the table below.

Carson's Reasoning	Jordyn's Reasoning																				
Carson said he noticed a pattern when looking at the table. Every time the number of days increased by 2, the cost increased by \$90. <table> <tr> <th>$x$</th><th>$y$</th></tr> <tr> <td>2</td><td>97</td></tr> <tr> <td>4</td><td>187</td></tr> <tr> <td>6</td><td>277</td></tr> <tr> <td>10</td><td>457</td></tr> </table> The y-intercept is the value of y when $x = 0$, so he used his pattern to work backwards. <table> <tr> <th>x</th><th>y</th></tr> <tr> <td>2</td><td>97</td></tr> <tr> <td>4</td><td>187</td></tr> <tr> <td>6</td><td>277</td></tr> <tr> <td>10</td><td>457</td></tr> </table>	x	y	2	97	4	187	6	277	10	457	x	y	2	97	4	187	6	277	10	457	Jordyn said that she can use the rate of change to find the y-intercept. She reasoned that if the daily rate is \$45 each day to rent the car, then the rate for 2 days should be \$90 and \$180 for 4 days. The difference between the daily rate and the actual cost represented in the table must be because of the maintenance fees.
x	y																				
2	97																				
4	187																				
6	277																				
10	457																				
x	y																				
2	97																				
4	187																				
6	277																				
10	457																				

- Use either student's reasoning to determine the y -intercept.
7
- What does this y -intercept represent in the context?
The maintenance fees are a flat \$7 charge.

QUESTIONING

Challenge students by asking them to compare and contrast how they find and interpret the slope and y -intercept with the four different representations (graph, equation, table, and written description). This analysis will enhance students' metacognitive awareness.

3. Madison planted a royal palm tree in her front yard in 2012. The tree was 12 feet tall when she planted it and has been growing at a nearly constant rate. By 2020, the royal palm measured 26 feet tall.
- Which two quantities are being compared in this linear relationship?
The height of the tree, in feet, and time, in number of years since 2012
 - Use the information to determine the slope and y -intercept in this relationship. Then, interpret each value given the context.
(0,12) (8,26)

Feature	Value	Interpretation
rate of change	1.75	<i>The tree's height increased by 1.75 feet each year since 2012.</i>
y -intercept	12	<i>The tree was 12 feet tall when planted in 2012.</i>



7.7.3 Guided Practice: Interpreting Slope and y-intercept from Tables and Descriptions

1. When landing, airplanes typically begin their descent at 37,000 feet above the ground and decrease altitude at a constant rate until they reach 10,000 feet above the ground. This initial descent takes about 9 minutes.

Use the information to determine the slope and y-intercept in this relationship. Then, interpret each value given the context.

Feature	Value	Interpretation
slope	-3,000	Planes descend about 3,000 feet per minute on their initial descent.
y-intercept	37,000	Planes are 37,000 feet above the ground when they start descending.

$(0, 37000) (9, 10000) \quad \frac{37,000 - 10,000}{0 - 9} = \frac{27,000}{-9} = -3,000$

2. A scuba diver exploring a coral reef begins her ascent to the surface of the water. The table models the relationship between the depth of the diver, d , in feet, and time, in minutes, t , after she begins ascending to the surface. Use the information to determine the slope and y-intercept in this relationship. Then, interpret each value given the context.

t	d
0	-130
2	-74
4	-18

Feature	Value	Interpretation
slope	28	She ascends 28 feet per minute.
y-intercept	-130	The diver is 130 feet below sea level when she starts swimming toward the surface.

$(0, -130) (2, -74) \quad \frac{-74 - (-130)}{2 - 0} = \frac{56}{2} = 28$

7.7.3 GUIDED PRACTICE: INTERPRETING SLOPE AND Y-INTERCEPT FROM TABLES AND DESCRIPTIONS

TEACHER MOVES

Again, allow students the opportunity to read the contexts on their own and discuss what is happening in each with their partner. Ensure students know the meaning of descend/descent and ascend/ascent.

ENRICHMENT

- Consider asking students the following:
- How can you determine the y-intercept from the written description?
 - How can you determine the slope with the information given?

QUESTIONING

Consider adding an error analysis question at the end of the activity to challenge student understanding of interpreting slope and y-intercept using tables and written descriptions in context.

7.7.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider allowing students to work with a partner or small group if needed after the 7.7.3 Guided Practice. Discussing the questions will build confidence and enhance student understanding.

ENRICHMENT

For each context:

- What are the quantities being compared?
- How can you determine the slope from the information given?
- How can you determine the y-intercept from the information given?

TEACHER MOVES

Consider facilitating a class discussion after question 2 to solidify the answers and address any student misconceptions.



7.7.4 Your Turn!

- Visitors to Ichetucknee Springs State Park in Fort White, Florida can rent tubes to float along the spring toward the South Takeout Point. Grayson begins his float 3.5 miles from the South Takeout Point. After 2 hours, he is 1.1 miles from the South Takeout Point. $(0, 3.5)$ $(2, 1.1)$

Use the information to determine the rate of change and y-intercept in this relationship. Then, interpret each value in context.

$$\frac{1.1 - 3.5}{2 - 0} = \frac{-2.4}{2} = -1.2$$

Feature	Value	Interpretation
rate of change	-1.2	Grayson gets 1.2 miles closer to the South Takeout Point every hour he floats.
y-intercept	3.5	Grayson is 3.5 miles from the South Takeout Point when he starts.

- Elijah is saving money to buy a new PlayStation. He set some money he received for his birthday aside and then added more each week. The table shows the amount of money Elijah has saved at the end of each week, m , for the number of weeks he has been saving, w .

w	1	2	5	7
m	80	110	200	260

Use the information to determine the rate of change and y-intercept in this relationship. Then, interpret each value in context.

$$\frac{110 - 80}{2 - 1} = \frac{30}{1} = 30 \quad 80 - 30 = 50$$

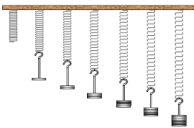
Feature	Value	Interpretation
rate of change	30	Elijah adds \$30 to his savings each week.
y-intercept	50	Elijah set aside \$50 of his birthday money to start.



7.7.5 Wrap-Up: Ticket Out the Door

In science class, Aliyah observed a linear relationship between the length a spring stretches and how it changes when different weights are attached to the end.

She observed the spring measures 3.7 centimeters (cm) when no weight is attached to it. When a 50 gram (g) weight is attached, the length of the spring measures 5.2 cm.



1. Use the information to determine the rate of change and y -intercept in this relationship. Then, interpret each value in context.

$(0, 3.7)$ $(50, 5.2)$ $\frac{5.2 - 3.7}{50 - 0} = \frac{1.5}{50} = 0.03$

Feature	Value	Interpretation
rate of change	0.03	The spring extends by 0.03 cm for every 1 gram of weight attached to it.
y -intercept	3.7	The spring is 3.7 cm long when no weight is attached.

7.7.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Discuss any questions the students still have regarding the content. Consider making a list of the questions on the board or poster to come back to and address later.